

SimpleScalar Computer Architecture Lab Experiments

Computer Architecture Laboratory

This folder contains the student experiment handouts for the SimpleScalar computer architecture lab sequence. Each experiment is provided as a PDF for students and as a LaTeX source file for editing. Repository layout after cloning:

```
SimpleScalar/  
  benchmarks/  
  config/  
  labs/  
  scripts/  
SimpleScalar/  
  experiments/  
  tests-alpha/  
  tests-pisa/  
  sim-fast.c  
  sim-safe.c  
  sim-profile.c  
  sim-bpred.c  
  sim-outorder.c  
  sim-singlecycle.c  
  Makefile
```

There is no nested `simplesim-3.0/` directory in this repository. Unless a lab explicitly says to enter `labs/`, run commands from the cloned repository root:

```
cd /path/to/SimpleScalar
```

1 Suggested Lab Sequence

Lab	Handout	Main topic	Simulator
0	00_setup_and_orientation.pdf	Setup, build targets, simulator overview	all
1	01_isa_profiling_lab.pdf	Instruction set profiling and Alpha vs PISA comparison	sim-profile
2	02_processor_design_lab.pdf	In-order and out-of-order pipelined processors	sim-outorder
2 extension	03_processor_design_home_assignment.pdf	Forwarding bottleneck and design-space analysis	sim-outorder
3	04_branch_prediction_lab.pdf	Branch predictor accuracy and CPI impact	sim-bpred, sim-outorder
4	05_singlecycle_processor_lab.pdf	Instructional single-cycle processor model	sim-singlecycle

Lab	Handout	Main topic	Simulator
5	06_compiling_custom_c_for_pisa.pdf	Compile custom C programs for PISA	sim-safe, sim-profile

2 Notes

- Lab 1 uses Alpha and PISA builds. Rebuild with `make clean` before switching ISA targets.
- Labs 2, 4, and 5 use PISA.
- Lab 3 uses Alpha benchmarks for branch prediction, matching the original UMass SimpleScalar reference lab.
- The local PISA cross compiler is installed under `toolchain/` by `scripts/install_pisa_cross_compiler.sh`, it is not committed to the repository.